MSSF-Net: A Multimodal Spectral–Spatial Feature Fusion Network for Hyperspectral Unmixing

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***Abstract*— Hyperspectral unmixing (HU) aims to decompose mixed pixels in remote sensing imagery into material-specific spectra and their respective abundance fractions. Recently, autoencoders (AEs) have made significant advances in HU due to their strong representational capabilities and ease of implementation. However, relying exclusively on feature extraction from a single-modality hyperspectral image (HSI) can fail to fully utilize both spatial and spectral information, thereby limiting the ability to distinguish objects in complex scenes. To address these limitations, we propose a multimodal spectral–spatial feature fusion network (MSSF-Net) for enhanced HU. The MSSF-Net adopts a dual-stream architecture to extract feature representations from complementary input modalities. Specifically, the hyperspectral subnetwork leverages a convolutional neural network (CNN) to capture spatial information, while the light detection and ranging (LiDAR) subnetwork incorporates an enhanced channel attention mechanism (ECAM) to capture the dynamic changes in spatial information across different channels. Furthermore, we introduce a cross-modal fusion (CMF) module that integrates spectral and spatial information across modalities, leading to more robust feature representations. Experimental results indicate that the MSSF-Net significantly outperforms existing traditional and deep learning (DL)-based methods in terms of unmixing accuracy. The code is available at https://github.com/Octopus-Squidward/MSSF-Net.**

***Index Terms*— Attention, autoencoder (AE), deep learning (DL), hyperspectral unmixing (HU), multimodal remote sensing image (MRSI).**

I. Introduction

HYPERSPECTRAL images (HSIs) are a valuable resource that combines imaging spectroscopy with remote sensing to capture both complex spatial characteristics and detailed spectral information of ground materials across hundreds of narrow and continuous bands, spanning visible to shortwave infrared (SWIR) wavelength regions. Known for its ability to capture extensive spatial–spectral information across multiple bands, HSI data have been widely applied in fields such as medical diagnostics [1], food safety [2], and military reconnaissance [3]. However, due to limitations in the spatial resolution of imaging spectrometers and the intricate distribution of natural surfaces, HSIs inevitably contain mixed pixels composed of multiple materials [4]. This mixing effect significantly impedes the accuracy of HSI processing in practical applications. Consequently, hyperspectral unmixing (HU) has become a critical task, aiming to decompose mixed pixels into distinct spectral signatures (endmembers) and estimate their corresponding fractions (abundance) [5].

Existing HU methods are typically categorized into linear and nonlinear models [6]. The linear mixing model (LMM) [7] assumes each pixel's spectrum results from a linear combination of endmember spectra, where photons interact with only one material before reaching the sensor, and each endmember is weighted by its abundance. However, in many real-world scenarios, photons undergo multiple scattering events and complex reflections, resulting in significant nonlinear mixing effects [8]. To better accommodate these real-world complexities, various non-LMMs (NLMMs) [9] have been developed to simulate photon scattering and surface reflections in more intricate scenarios [10], [11], [12].

Over the past decade, deep neural networks have made substantial progress in feature representation, leading to their increasing application in HU [13], [14]. Among these, the autoencoder (AE) has garnered considerable attention for its effective unmixing capabilities. The primary objective of an AE is to minimize reconstruction error, thereby learning optimal low-dimensional representations of the input data [15], [16]. Typically, the AE maps HSI data to abundance estimates while reconstructing high-dimensional representations. More recently, the AE has undergone various enhancements, with optimizations focusing on sparsity [17], spatial–spectral characteristics [18], endmember variability [19], and denoising [20]. For instance, Qi et al. [21] proposed the spatial–spectral collaborative unmixing network (SSCU-Net), which integrates spatial and spectral AE networks with superpixel segmentation to improve unmixing accuracy through collaborative training and effective utilization of spatial information. The cycle-consistency unmixing network (CyCU-Net) [22] employs an end-to-end framework with two cascaded AEs to enforce cycle consistency, reducing detail and material information loss by relaxing the pixel-level reconstruction assumption. The probabilistic generative model for spectral unmixing (PGMSU) [23] introduces a probabilistic generative model to tackle endmember variability, utilizing the nonlinear modeling capability of neural networks to fit arbitrary endmember distributions.

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Similarly, the deep generative model for spatial–spectral unmixing (DGMSSU) [24] associates generated endmembers with the probability distributions of endmember bundles extracted from hyperspectral imagery. By incorporating discriminators alongside graph convolutional networks or self-attention modules, DGMSSU flexibly handles irregular data structures. On the other hand, the proportional perturbation model (PPM) [25] addresses endmember variability by introducing scaling factors and a perturbation term, enabling it to capture both proportional changes in endmembers and localized fluctuations in real-world scenarios. Qi et al. [26] introduced the multiview spatial–spectral two-stream network (MSSS-Net), which integrates multiview spectral and spatial information through spectral partitioning and recurrent neural networks (RNNs). By simultaneously processing spatial and spectral streams with cascaded RNN encoders, MSSS-Net enhances feature discrimination and improves unmixing efficiency. Similarly, Qi et al. [27] proposed the spatial attention-weighted unmixing network (SAWU-Net), which incorporates pixel and window attention blocks with dynamic abundance weighting to refine spatial feature extraction and enhance HU. Zheng et al. [28] developed a physics-driven two-stream AE network that integrates the dispersion model (DM) into the endmember extraction (EE) network, while sharing weights with the abundance estimation network, leading to improved endmember and abundance estimation. Additionally, Chen et al. [29] introduced the Transformer and CNN collaborative unmixing network (TCCU-Net), a dual-stream network that synergizes a Transformer encoder and a CNN encoder to enable end-to-end learning of spectral, spatial, global, and local features. HU with the multiscale convolution attention network (HUMSCAN) [30] combines the endmember unmixing network (EU-Net) and the abundance unmixing network (AU-Net), allowing unsupervised extraction of endmembers and their corresponding abundance information simultaneously. The deep attention-guided spatial–spectral network (DASS-Net) [18] employs a dual-stream architecture incorporating the neighborhood spatial attention (NSA) module and a dual-gated mechanism, facilitating adaptive spatial–spectral feature extraction and enhancing unmixing performance. Despite the effectiveness of these AE-based models for HU, they primarily rely on single-modality input, which limits their ability to distinguish between objects made of similar materials. To overcome this limitation, it is necessary to develop and integrate multisource data to further enhance the accuracy and robustness of HU.

With the rapid advancement of sensor technology, various types of remote sensing data can capture distinct properties of the same Earth's surface. In addition to HSIs, light detection and ranging (LiDAR) offers ground elevation data [31], [32], while synthetic aperture radar (SAR) provides structural information about the surface [33], [34]. Consequently, the fusion of HSI and LiDAR data has been extensively studied in remote sensing image classification. For instance, Zhao et al. [35] employed a hierarchical random walk network (HRWN) with a dual-tunnel convolutional neural network (CNN) and a pixel-wise affinity branch for joint classification of hyperspectral imagery and LiDAR data. Zhang et al. [36] developed a three-branch CNN model to merge spectral, spatial, and elevation data into multilevel fusion features (MLFs). Shen et al. [37] proposed the shuffle feature-enhanced fusion network (SFEFN), which facilitates multimodal exchanges between HSIs and LiDAR, reducing feature discrepancies. Feng et al. [38] introduced a self-supervised cross-modal contrastive learning (CMCL) method with hybrid cross-modal fusion (CMF) modules for robust feature representation in multimodal remote sensing image (MRSI) classification. Additionally, Li et al. [39] presented an innovative HyperMLP network based on the MLP-mixer architecture to enhance spatial feature construction and multimodal feature fusion, leveraging unsupervised superpixel embedding, feature scrambling, and bilateral modulation strategies. In contrast, fewer studies have explored the use of LiDAR data in HU. Uezato et al. [40] developed a general HU framework that integrates LiDAR data to guide spatial regularization. Subsequently, to capture spatial correlations among neighboring pixels, a hypergraph regularization is introduced to optimize and enhance abundance estimation [41]. Han et al. [42] proposed a multimodal unmixing network (MUNet) incorporating squeeze-and-excitation attention and AE architectures to fuse HSI and LiDAR features, improving abundance estimation. Hu et al. [43] introduced a spectral–spatial fusion attention network (SFANet), which integrates the exponential local attention module (ELAM) and dynamic group convolutions to fuse HSI and LiDAR data, thereby improving the accuracy of spectral unmixing. Although these HU methods incorporate spatial regularization and spectral information, they fall short of fully exploiting joint spectral–spatial contexts or effectively integrating HSI and LiDAR features.

To address these challenges, we propose a multimodal spectral–spatial feature fusion network, called MSSF-Net. This network integrates 2-D convolution and an enhanced channel attention mechanism (ECAM) [44] in the AE unmixing framework, enabling unsupervised feature extraction from both HSI and LiDAR data. Additionally, a CMF module is incorporated to further integrate heterogeneous features. Compared to existing multimodal methods that rely solely on spectral information, the MSSF-Net achieves a more comprehensive integration of cross-modal spatial and spectral information, leading to more efficient and accurate abundance estimation. Specifically, the main contributions of this article are summarized as follows.

1) We propose a MUNet, MSSF-Net, designed to effectively capture and integrate rich spatial–spectral information from multisource data, thereby improving both the accuracy and reliability of HU. This study aims to enhance the efficiency and comprehensiveness of spectral–spatial feature fusion, providing more effective guidance for the unmixing process.
2) A dual-stream feature extraction module is introduced to overcome the limitations in utilizing spatial–spectral information. By combining CNNs with ECAM, this module efficiently extracts relevant spatial–spectral features and integrates them into the MSSF-Net, thereby

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enhancing its capability to capture and fuse features across different modalities.

3) We propose a CMF module to improve the fusion effectiveness of multimodal features. This module precisely integrates local detail features from multimodal data while consolidating global information, significantly boosting the model's unmixing accuracy.

The rest of this article is organized as follows. Section II provides an in-depth review of related work on unmixing techniques, AE theory, and multimodal data unmixing strategies. Section III elaborates on the design and implementation of the MSSF-Net framework. Section IV presents a series of carefully designed experiments that comprehensively validate the effectiveness and superiority of the MSSF-Net on both synthetic and real-world datasets. Finally, Section V concludes the article.

II. Related Work

In this section, we provide an overview of existing multimodal and AE-based unmixing frameworks, followed by a discussion of the CMF module.

A. Existing Linear Multimodal Mixing Model

Under the assumption of the LMM [7], each observed pixel is a convex combination of endmembers, weighted by their corresponding fractional abundances. Let $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N] \in \mathbb{R}^{L\times N}$ represent the hyperspectral data, where $\mathbf{y}_n$ denotes the nth observed spectrum, L is the number of spectral bands, and N is the total number of HSI pixels. The LMM for HSIs can be formulated as

$$\mathbf{Y} = \mathbf{E}\mathbf{A} + \mathbf{N} \tag{1}$$

where $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_P] \in \mathbb{R}^{L\times P}$ is the endmember matrix, with P being the number of endmembers and $\mathbf{e}_p$ the pth endmember. The matrix $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_N] \in \mathbb{R}^{P\times N}$ represents the abundance matrix, where $\mathbf{a}_n$ corresponds to the abundance vector for the nth pixel. Since the abundances represent the relative fractions of each material, they are required to satisfy the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC), which are given by

$$\begin{cases} \mathbf{a}_n \geq \mathbf{0} \\ \mathbf{1}^\top \mathbf{a}_n = 1 \end{cases} \tag{2}$$

where $\mathbf{0}$ and $\mathbf{1}$ denote $P \times 1$ column vectors with all entries equal to zero and one, respectively. Furthermore, endmembers must satisfy the endmember nonnegativity constraint (ENC), that is, $\mathbf{e}_p \geq \mathbf{0}$ for $1 \leq p \leq P$, because their characteristic shapes depend on the material's reflectance or radiance [5]. Finally, $\mathbf{N}$ accounts for the measurement noise in the observation.

In multimodal unmixing frameworks, multiple data modalities, such as LiDAR and SAR, are incorporated to guide the extraction of more robust abundance representations through spatial regularization. Compared to existing single-modal methods, multimodal approaches leverage complementary features from different modalities to enhance unmixing accuracy. In this work, we use LiDAR data as an additional modality to complement HSIs, as LiDAR data provide height information, which helps differentiate spectrally similar but spatially distinct objects. This additional information improves the estimation of endmembers and abundance values, leading to more accurate unmixing.

B. Related AE-Based Model for Unmixing

Recently, AE-based techniques have shown strong potential in HU by simultaneously extracting both the abundance matrix $\mathbf{A}$ and the endmember matrix $\mathbf{E}$ in an unsupervised manner. The AE framework typically comprises two main components: an encoder and a decoder.

1) *Encoder:* The encoder is a multilayer network structure that learns the hidden representation $\mathbf{v}_n$ of the input raw multimodal data, which is defined by

$$\mathbf{v}_n = f_E(\mathbf{y}_n) = f\left(\left(\mathbf{W}^{(i)}\right)^\top \mathbf{y}_n + \mathbf{b}^{(i)}\right) \tag{3}$$

where the superscript $(\cdot)^\top$ denotes the transpose of a matrix, $f(\cdot)$ represents the nonlinear activation function of the encoder, $\mathbf{W}^{(i)}$ denotes the weights of the ith encoder layer, and $\mathbf{b}^{(i)}$ is the bias of the ith encoder layer.

2) *Decoder:* According to the definition of LMM, the decoder's goal is to reconstruct the input pixel $\mathbf{y}_n$ from the extracted hidden representation $\mathbf{v}_n$, which can be expressed as

$$\hat{\mathbf{y}}_n = f_D(\mathbf{v}_n) = \left(\mathbf{W}^{(j)}\right)^\top \mathbf{v}_n \tag{4}$$

where $\mathbf{W}^{(j)}$ denotes the weights of the decoder and $\hat{\mathbf{y}}_n \in \mathbb{R}^L$ is the reconstructed pixel. Since the decoder function aligns with the characteristics of the LMM, it can be interpreted that the hidden representation output by the encoder corresponds to the estimated abundance vector $\hat{\mathbf{a}}_n$, and the weights of the decoder represent the estimated endmember matrix $\widehat{\mathbf{E}}$, which are defined by

$$\mathbf{v}_n \to \hat{\mathbf{a}}_n \tag{5}$$

$$\mathbf{W} \to \widehat{\mathbf{E}}. \tag{6}$$

Therefore, the AE framework efficiently performs HU by decomposing the input data into abundance and endmember matrices, significantly enhancing precision and computational efficiency.

C. Multimodal Image Feature Fusion Methods

HSIs provide rich spectral information; however, relying solely on them may not yield optimal results in unmixing tasks, particularly when dealing with spectrally similar or spatially adjacent objects. Recent studies have shown that fusing features from HSI and LiDAR data significantly enhances unmixing accuracy [40], [42]. Feature fusion has thus become a key approach in most research efforts. Within this context, an effective CMF module is critical for optimizing the complementary information from HSI and LiDAR data. While traditional methods have their strengths, deep learning (DL) has emerged as a powerful tool for handling

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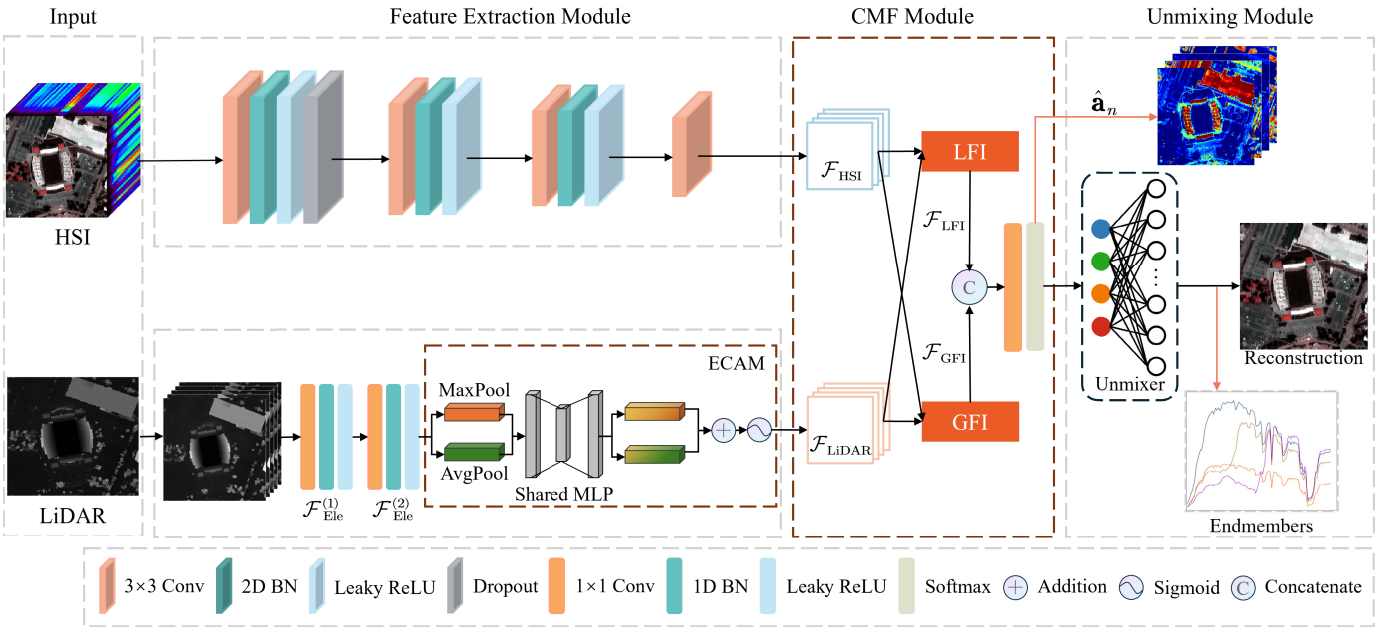


Fig. 1. Architecture of the proposed MSSF-Net framework, featuring a dual-stream multimodal feature extraction module composed of HSI and LiDAR feature extraction branches. The heterogeneous features are integrated through the CMF module and subsequently fed into the decoder.

complex multimodal data. CNN-based models, in particular, are instrumental in fusing HSI and LiDAR features. For instance, Han et al. [45] proposed the cross-modal semantic enhancement (CMSE) network, which effectively mines and integrates multimodal semantic information, reducing feature discrepancies between the two data types. Similarly, with the integration of DL and attention mechanisms, multimodal data fusion has become even more efficient. FusAtNet [46], an attention-based multimodal fusion network for land-cover classification, exemplifies this by leveraging both spectral and spatial information. These advanced methods capitalize on the complementary strengths of HSI and LiDAR data, enabling the development of more robust representations of complex scenes. As a result, they not only enhance unmixing accuracy but also improve resilience across a wide range of applications.

III. Proposed Method

In this section, we present the MSSF-Net framework for HU. The overall architecture of the network is illustrated in Fig. 1. First, a multimodal feature extraction module is employed to capture rich spatial and spectral features from both HSI and LiDAR data. Next, the CMF module integrates local features with global contextual information to enhance feature representation. Finally, the reconstructed results are generated through a series of linear layer operations.

A. Multimodal Feature Extraction Module

To fully exploit the complementary information from HSI and LiDAR data, we propose a dual-branch multimodal feature extraction module. This module consists of two components: the HSI feature extraction module and the LiDAR feature extraction module, which encode data from their respective modalities to capture both spectral and spatial features effectively.

1) HSI Feature Extraction Module: In the HSI feature extraction module, spatial features are extracted using four consecutive 3×3 convolutional blocks. Unlike standard 1×1 convolutions, which primarily focus on spectral information, 3×3 convolutions are employed to capture spatial details more effectively. Following each convolution operation, activation functions such as Tanh, ReLU, RReLU, or Leaky ReLU are applied [47]. The ReLU function sets negative input values to zero, potentially leading to neuron inactivity. To mitigate this issue, Leaky ReLU is adopted as it allows small gradients for negative inputs. Furthermore, batch normalization (BN) is incorporated to address the vanishing gradient problem and improve training stability.

2) LiDAR Feature Extraction Module: The LiDAR feature extraction module preprocesses the LiDAR image $\mathbf{I}$ by extending it into a multiband profile using attribute profiles (AP) [48], [49]

$$\mathrm{AP}_{\mathbf{I}} = \{\varphi_M(\mathbf{I}), \ldots, \varphi_1(\mathbf{I}), \mathbf{I}, \rho_1(\mathbf{I}), \ldots, \rho_M(\mathbf{I})\} \tag{7}$$

where $\mathrm{AP}_{\mathbf{I}} \in \mathbb{R}^{H_{\mathbf{I}} \times W_{\mathbf{I}} \times (2M+1)}$ represents the AP-transformed LiDAR data, and $H_{\mathbf{I}}$ and $W_{\mathbf{I}}$ denote to the height and width of the LiDAR image $\mathbf{I}$, respectively. Here, φ_m and ρ_m correspond to the attribute thinning and thickening transformations, respectively. During the attribute thinning process, the image $\mathbf{I}$ is thresholded at each gray level, followed by binary attribute thinning filtering for further refinement. This process enhances the boundaries between major surface features and improves the differentiation between materials. The final output is determined by the maximum gray level attained for each pixel during filtering. Conversely, the attribute thickening transformation is derived from the minimum gray level of the results obtained after filtering for each pixel, effectively filling small gaps within objects and enriching the representation of surface morphology. After applying thinning and thickening

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transformations to the LiDAR data, the elevation information is significantly optimized, generating a refined dataset. By concatenating these processed data with the original LiDAR data, new spatial feature dimensions are introduced, effectively expanding the feature space and improving the overall representation of terrain variations.

Subsequently, two consecutive 1×1 convolution layers are applied to compress feature information and extract the elevation features $\mathcal{F}_{\text{Ele}}^{(2)} \in \mathbb{R}^{B\times P\times 1\times 1}$ from $\text{AP}_{\mathbf{I}}$, where B is the batch size of the input image patch. To further enhance feature extraction, the ECAM dynamically adjusts the importance of each channel, enabling the model to prioritize essential information effectively. This process begins by applying global max-pooling and global average pooling to the feature map $\mathcal{F}_{\text{Ele}}^{(2)}$ obtained from the second convolutional layer of the LiDAR feature extraction module. These pooling operations generate two feature maps of size $P \times 1 \times 1$. The pooled feature maps are then passed through a shared multilayer perceptron (MLP), and their outputs are summed element-wise. The resulting values are activated using a sigmoid function to produce the elevation attention map $\mathbf{M}_{\text{Ele}}$, which is defined as

$$\mathbf{M}_{\text{Ele}} = \sigma\left(\text{MLP}\left(\text{Max}\left(\mathcal{F}_{\text{Ele}}^{(2)}\right)\right) \oplus \text{MLP}\left(\text{Avg}\left(\mathcal{F}_{\text{Ele}}^{(2)}\right)\right)\right) \quad (8)$$

where $\sigma(\cdot)$ is the sigmoid activation function, $\oplus$ denotes element-wise addition, and $\text{Max}(\cdot)$ and $\text{Avg}(\cdot)$ denote global max-pooling and global average pooling, respectively. This mechanism dynamically captures critical spatial variations in the LiDAR data expanded by the AP technique, thereby reinforcing the model's ability to extract spatial features and improving the efficiency of multimodal feature extraction. By adaptively weighting channel importance, ECAM ensures that the most relevant elevation information is prioritized, contributing significantly to the overall performance of the model.

B. CMF Module

Effectively capturing and integrating spectral and spatial information from different modalities is critical for achieving accurate unmixing in HSIs. Traditional approaches, such as simple weighting or summing of features from two modalities, often fail to fully exploit their intricate interactions, thereby limiting the precision of the unmixing process. To overcome this limitation, we propose a novel CMF module that incorporates two key components: the local feature interaction (LFI) component and the global feature interaction (GFI) component. The LFI component focuses on local feature integration by employing feature concatenation techniques to merge multimodal features within localized regions, facilitating an initial fusion at the local level. In contrast, the GFI component operates from a broader perspective, leveraging an interactive cross-modal attention mechanism to deeply analyze and establish relationships between features from different modalities. This enables a more comprehensive exploration of dependencies among features at a global scale. By combining the complementary strengths of the LFI and GFI components, the CMF module achieves a more comprehensive and effective integration of multimodal features. This synergistic approach lays a robust foundation for subsequent data analysis, significantly enhancing the accuracy and reliability of the unmixing process.

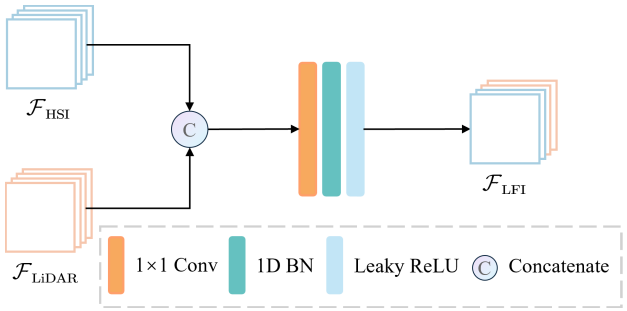


Fig. 2. Detailed illustration of the LFI component.

1) LFI Components: The outputs of the hyperspectral feature extraction module and LiDAR feature extraction module are denoted as $\mathcal{F}_{\text{HSI}} \in \mathbb{R}^{B\times P\times 1\times 1}$ and $\mathcal{F}_{\text{LiDAR}} \in \mathbb{R}^{B\times P\times 1\times 1}$, respectively. Although derived from distinct modalities, these features contain complementary information at corresponding spatial locations. To integrate these cross-modal features, we first concatenate $\mathcal{F}_{\text{HSI}}$ and $\mathcal{F}_{\text{LiDAR}}$ along the channel dimension, which is given by

$$\mathcal{F}_{\text{Con}} = \text{Concat}(\mathcal{F}_{\text{HSI}}, \mathcal{F}_{\text{LiDAR}}) \quad (9)$$

where $\text{Concat}(\cdot,\cdot)$ denotes the concatenation operation and $\mathcal{F}_{\text{Con}} \in \mathbb{R}^{B\times 2P\times 1}$. Next, a convolution operation with a 1×1 kernel is applied to achieve deep fusion and efficient utilization of these features, as follows:

$$\mathcal{F}_{\text{LFI}} = \text{LeakyReLU}(\text{BN}(f_{1\times 1}(\mathcal{F}_{\text{Con}}))) \quad (10)$$

where $\mathcal{F}_{\text{LFI}} \in \mathbb{R}^{B\times P\times 1\times 1}$, and $\text{LeakyReLU}(\cdot)$ denotes the activation function

$$\text{LeakyReLU}(x) = \begin{cases} x, & x \geq 0 \\ \alpha x, & x < 0 \end{cases} \quad (11)$$

where α is set to a default value of 0.01 in the following experiments. The detailed architecture of the LFI component is illustrated in Fig. 2.

2) GFI Components: The GFI component incorporates an interactive cross-modal attention mechanism that enables deep feature extraction and efficient fusion of HSI and LiDAR information. This mechanism fully exploits the complementary nature of the two modalities, while the fusion strategy further enhances unmixing accuracy and model generalization. The detailed architecture of the GFI component is illustrated in Fig. 3. The process begins by applying a 1×1 convolution to the inputs feature maps, $\mathcal{F}_{\text{HSI}}$ and $\mathcal{F}_{\text{LiDAR}}$, followed by a squeeze operation to extract key attention values

$$\mathbf{F}_{\text{HSI}} = \text{Squeeze}(f_{1\times 1}(\mathcal{F}_{\text{HSI}})) \in \mathbb{R}^{B\times P} \quad (12)$$

$$\mathbf{F}_{\text{LiDAR}} = \text{Squeeze}(f_{1\times 1}(\mathcal{F}_{\text{LiDAR}})) \in \mathbb{R}^{B\times P} \quad (13)$$

where $\text{Squeeze}(\cdot)$ denotes a channel-wise squeeze operation, which removes singleton dimensions to produce a more compact feature representation. The 1×1 convolutional layer enhances the discriminability of the original feature maps by

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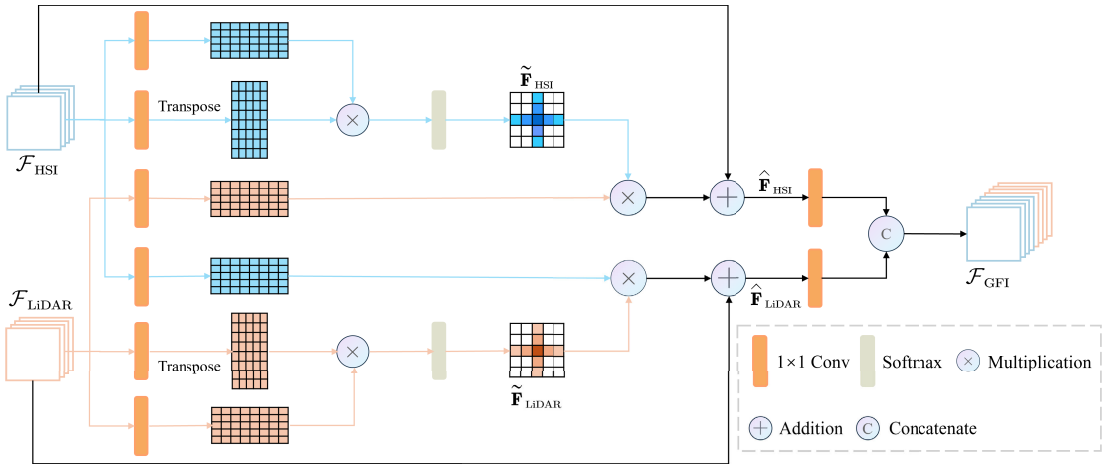


Fig. 3. Graphical overview and computational flow of the GFI component.

extracting essential features. Next, an attention mechanism is applied to the HSI and LiDAR feature maps, $\mathbf{F}_{\text{HSI}}$ and $\mathbf{F}_{\text{LiDAR}}$, to compute their respective attention weights

$$\widetilde{\mathbf{F}}_{\text{HSI}} = \text{softmax}\left(\mathbf{F}_{\text{HSI}}^{\top} \cdot \mathbf{F}_{\text{HSI}}\right) \in \mathbb{R}^{P\times P} \tag{14}$$

$$\widetilde{\mathbf{F}}_{\text{LiDAR}} = \text{softmax}\left(\mathbf{F}_{\text{LiDAR}}^{\top} \cdot \mathbf{F}_{\text{LiDAR}}\right) \in \mathbb{R}^{P\times P}. \tag{15}$$

The GFI component then establishes cross-modal attention between HSI and LiDAR, facilitating comprehensive global interaction. Specifically, it captures the internal importance of each modality while precisely identifying cross-modal dependencies. The attention-refined features are computed as follows:

$$\widehat{\mathbf{F}}_{\text{HSI}} = \left(\mathbf{F}_{\text{LiDAR}} \cdot \widetilde{\mathbf{F}}_{\text{HSI}}\right) \oplus \text{Squeeze}(\mathcal{F}_{\text{HSI}}) \tag{16}$$

$$\widehat{\mathbf{F}}_{\text{LiDAR}} = \left(\mathbf{F}_{\text{HSI}} \cdot \widetilde{\mathbf{F}}_{\text{LiDAR}}\right) \oplus \text{Squeeze}(\mathcal{F}_{\text{LiDAR}}) \tag{17}$$

where $\widehat{\mathbf{F}}_{\text{HSI}} \in \mathbb{R}^{B\times P}$ and $\widehat{\mathbf{F}}_{\text{LiDAR}} \in \mathbb{R}^{B\times P}$. By applying attention weights from each modality to the other's key features, the model establishes bidirectional cross-modal interactions, generating enhanced feature representations. To further improve feature robustness and discriminability, an additional 1×1 convolution layer is applied. The cross-attention refined features are then concatenated to form a global comprehensive representation

$$\mathbf{F}_{\text{GFI}} = \text{Concat}\left(f_{1\times1}\left(\widehat{\mathbf{F}}_{\text{HSI}}\right), f_{1\times1}\left(\widehat{\mathbf{F}}_{\text{LiDAR}}\right)\right) \in \mathbb{R}^{B\times 2P}. \tag{18}$$

Finally, to ensure compatibility with subsequent processing stages, the dimensions of $\mathbf{F}_{\text{GFI}}$ are expanded to $\mathcal{F}_{\text{GFI}} \in \mathbb{R}^{B\times 2P\times 1\times 1}$. This final feature map $\mathcal{F}_{\text{GFI}}$ integrates complementary information from both modalities, enabling comprehensive spectral–spatial feature extraction and fusion.

The network configuration of the proposed MSSF-Net is summarized in Table I. The architecture is divided into seven blocks: blocks 1–4 are dedicated to multimodal feature extraction, blocks 5 and 6 handle cross-modal feature fusion, and block 7 serves as the decoder.

TABLE I

NEURAL NETWORK CONFIGURATION OF THE MSSF-NET. "CONV" DENOTES CONVOLUTIONAL OPERATION

Architecture	HSI		LiDAR	
	Layers	Unit	Layers	Unit
Multimodal Feature Extraction Module	3×3 Conv BN LeakyReLU	$32P$	1×1 Conv BN LeakyReLU	$\frac{2M+1}{2}$
	3×3 Conv BN LeakyReLU	$16P$	1×1 Conv BN LeakyReLU	P
	3×3 Conv BN LeakyReLU	$4P$	ECAM	P
	3×3 Conv	P		
CMF Module	Layers	LFI	GFI	Concat
	Unit	P	$2P$	$3P$
Decoder	Layers		Unit	
	1×1 Conv softmax		P	

C. Unmixing Module

In the CMF module, to derive more robust complementary fusion features, the outputs of the LFI and GFI components are concatenated as follows:

$$\mathcal{F}_{\text{output}} = \text{softmax}(f_{3\times3}(\text{Concat}(\mathcal{F}_{\text{LFI}}, \mathcal{F}_{\text{GFI}}))) \tag{19}$$

where $f_{3\times3}(\cdot)$ denotes a convolutional layer with a 3×3 kernel and $\mathcal{F}_{\text{output}} \in \mathbb{R}^{B\times P\times 1\times 1}$. During the evaluation phase, the dimensions of $\mathcal{F}_{\text{output}}$ are reduced to $P \times B$, yielding the estimated abundance matrix $\widehat{\mathbf{A}}$, which is formulated as

$$\widehat{\mathbf{A}} = \text{Squeeze}\left(\mathcal{F}_{\text{output}}\right)^{\top}. \tag{20}$$

The concatenated features are subsequently passed through a fully connected layer in the unmixing module. The decoder reconstructs the pixels using the estimated abundance matrix and the corresponding endmember matrix as follows:

$$\hat{\mathbf{y}}_n = f_D\left(\left(\mathbf{W}^{(j)}\right)^{\top}\hat{\mathbf{a}}_n\right) = \widehat{\mathbf{E}}\,\hat{\mathbf{a}}_n \tag{21}$$

where $\hat{\mathbf{y}}_n \in \mathbb{R}^L$ denotes the reconstructed pixel and $f_D(\cdot)$ represents a notational identifier for the linear decoder.

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D. Loss Function

To optimize the performance of the MSSF-Net, the following loss function design is proposed. The primary loss component is the mean squared error (MSE), which minimizes the reconstruction error between the input pixels and the reconstructed pixels

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{n=1}^{N}\|\mathbf{y}_n - \hat{\mathbf{y}}_n\|_2^2 \tag{22}$$

where $\mathbf{y}_n$ and $\hat{\mathbf{y}}_n$ represent the input pixels and the reconstructed pixels, respectively.

Since the Softmax function does not inherently promote sparsity, we incorporate sparsity-enforcing constraints into the model to estimate abundances [50]

$$\mathcal{L}_{\text{Spr}} = \|\hat{\mathbf{A}}\|_{1/2} = \sum_{i=1}^{P}\sum_{j=1}^{N}\left|\hat{a}_{ij}\right|^{1/2}. \tag{23}$$

Compared to the ℓ_1 norm, which is less effective in promoting sparsity, and the ℓ_2 norm, which ensures smoothness but lacks the essential sparsity property, the $\ell_{1/2}$ norm provides a more effective sparsity constraint on the abundance matrix [51].

To further enhance endmember estimation accuracy, we incorporate the spectral angle distance (SAD)

$$\mathcal{L}_{\text{SAD}} = \frac{1}{LP}\sum_{i=1}^{P}\sum_{j=1}^{L}\arccos\frac{\bar{\mathbf{e}}_j^{\top}\hat{\mathbf{e}}_{ji}}{\|\bar{\mathbf{e}}_j\|_2\|\hat{\mathbf{e}}_{ji}\|_2} \tag{24}$$

where $\bar{\mathbf{e}}_j = (1/L)\sum_{i=1}^{L}\hat{\mathbf{e}}_{ji}$ represents the mean of the estimated endmember vectors.

Additionally, the minimum volume constraint (MVC) [52] has demonstrated promising potential in EE by promoting the formation of a compact simplex. To further quantify the compactness of the endmember simplex and ensure stable endmember estimation, we introduce endmember distance regularization [53], which is mathematically defined as

$$\mathcal{L}_{\text{MVC}} = \frac{1}{LP}\sum_{p=1}^{P}\left\|\hat{\mathbf{e}}_p - \frac{1}{P}\sum_{p=1}^{P}\hat{\mathbf{e}}_p\right\|_2^2. \tag{25}$$

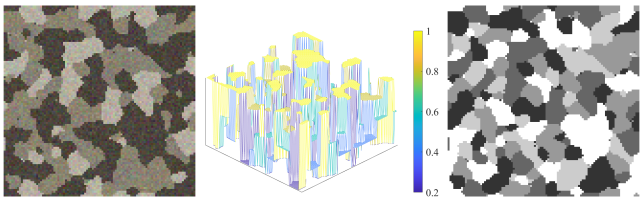
The final total loss function for MSSF-Net is defined as

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{Spr}}\mathcal{L}_{\text{Spr}} + \lambda_{\text{SAD}}\mathcal{L}_{\text{SAD}} + \lambda_{\text{MVC}}\mathcal{L}_{\text{MVC}} \tag{26}$$

where λ_{Spr}, λ_{SAD}, and λ_{MVC} are hyperparameters that control the contributions of each loss term.

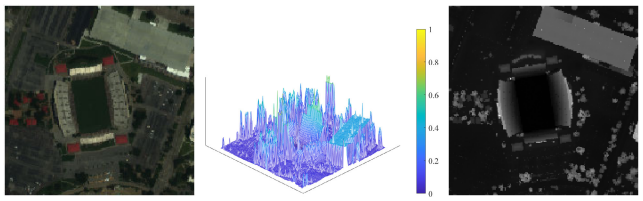
IV. Experimental Results

In this section, we present a comprehensive performance evaluation of the proposed method. The evaluation is conducted using a meticulously constructed synthetic dataset and two widely used real hyperspectral datasets to ensure robustness and reliability. To highlight the advantages and unique characteristics of the proposed approach, we compare it against eight state-of-the-art HU methods, categorized into three primary groups based on their core techniques. The first group includes traditional methods, represented by fully constrained least squares unmixing (FCLSU) [54]. The second group consists of CNN-based models, including CyCU-Net [22], PGMSU [23], DGMSSU [24], and HUMSCAN [30]. The third group features multimodal approaches, such as weighted spatial regularization from the digital surface model (w-DSM) [40], MUNet [42], and SFANet [43].



(a) (b) (c)

Fig. 4. Visualization of the SIM1 dataset. (a) Hyperspectral pseudo-color image. (b) Synthetic DSM data. (c) LiDAR grayscale image.



(a) (b) (c)

Fig. 5. Visualization of the Houston dataset. (a) Hyperspectral pseudo-color image. (b) Synthetic DSM data. (c) LiDAR grayscale image.

By incorporating a comprehensive selection of comparison methods, we systematically evaluate the performance of the proposed method in HU tasks. The evaluation focuses on several key aspects, including unmixing accuracy to quantify the precision of abundance estimations, computational efficiency to evaluate the practicality of the method for real-time or large-scale applications, and adaptability to examine its effectiveness across diverse datasets, particularly in complex real-world scenarios. This evaluation highlights the ability of the proposed method to address challenges in HU, such as spectral variability and spatial information fusion, demonstrating superior performance across both synthetic and real-world datasets when compared to existing methods.

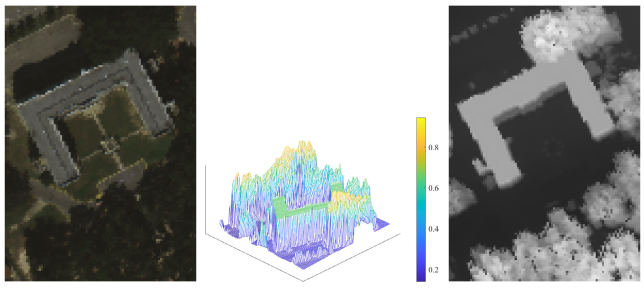
A. HSI Data

1) Synthetic Dataset: For evaluation, we use the synthetic dataset SIM1, as described in [40]. This dataset includes five endmember spectra selected from the United States Geological Survey (USGS) spectral library, with a spatial resolution of 100 × 100 pixels. Each pixel contains 224 spectral bands spanning the visible to near-infrared (VNIR) and SWIR regions. The hyperspectral data is generated using the LMM and visualized as an RGB image in Fig. 4(a). Additionally, the synthetic DSM is modeled by introducing Gaussian noise, resulting in a signal-to-noise ratio (SNR) of 50 dB, as shown in Fig. 4(b). Furthermore, a multiband LiDAR profile is synthesized using AP, as depicted in Fig. 4(c).

2) Houston Dataset: The Houston2013 dataset was introduced by the IEEE Geoscience and Remote Sensing Society (GRSS) for the 2013 data fusion contest.[1] Acquired in

[1]https://hyperspectral.ee.uh.edu/

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(a) (b) (c)

Fig. 6. Visualization of the Muffle dataset. (a) Hyperspectral pseudo-color image. (b) Synthetic DSM data. (c) LiDAR grayscale image.

2012 using the ITRES CASI-1500 sensor, the dataset covers the University of Houston campus and its surroundings with a spatial resolution of 2.5 m. It spans a wavelength range of 0.38–1.05 μm, providing 144 spectral bands rich in information for spectral analysis. For this study, a 170 × 170 pixel subimage was extracted from the larger original 349 × 1905 pixel image to focus on a specific region for detailed analysis, as shown in Fig. 5(a). To enhance the experimental comprehensiveness, the corresponding DSM and LiDAR data for this subimage were included as auxiliary inputs. Fig. 5(b) and (c), respectively, displays the detailed DSM and LiDAR images. This selected region features four key endmembers: #1 Parking Lot1, #2 Parking Lot2, #3 Running Track, and #4 Grass Healthy.

3) Muffle Dataset: Collected in 2010 on the University of Mississippi campus [55], the Muffle dataset[2] offers a spatial resolution of 1 m, enabling accurate capture of fine ground details. It comprises 64 continuous spectral bands within a wavelength range of 0.375–1.050 μm, providing abundant spectral information. From the original 325 × 220 pixel dataset, a representative subimage of 130 × 90 pixels was extracted for analysis, as shown in the RGB image in Fig. 6(a). To further enrich this dataset, DSM and LiDAR data were incorporated as auxiliary inputs. Fig. 6(b) presents the DSM image, highlighting surface topography variations, while Fig. 6(c) shows the LiDAR image, adding a multisource layer of information. The analysis focuses on five key material categories: #1 Roof, #2 Grass, #3 Tree, #4 Shadow, and #5 Asphalt.

B. Experimental Setup

1) Hyperparameter Settings: The MSSF-Net model employed in this study is implemented using the PyTorch framework to efficiently address complex HU tasks. Training is conducted on a high-performance hardware setup comprising an i5-13600KF CPU and an RTX 4070 SUPER 12G GPU. The Adam optimizer is utilized for parameter updates, ensuring effective and stable convergence. To balance training efficiency with model generalization, the batch size is set to 256 for the Muffle dataset and 128 for the other datasets. Hyperparameter tuning is a critical component of the training process, with optimized settings tailored to each dataset, as detailed in Table II.

[2]https://github.com/GatorSense/MUUFLGulfport

TABLE II
NETWORK HYPERPARAMETER CONFIGURATION OF THE MSSF-NET

Dataset	Learning Rate	λ_{Spr}	λ_{SAD}	λ_{MVC}	Batch Size
SIM1	7×10^{-3}	8×10^{-4}	1	5×10^{-2}	128
Houston	1×10^{-2}	1×10^{-3}	0.5	5	128
Muffle	1×10^{-2}	8×10^{-4}	5	0.6	256

2) Quantitative Performance Measures: In the experiments, the unmixing performance of the proposed method was comprehensively evaluated using the root mean square error (RMSE) for abundance estimates across all endmembers, along with the average RMSE (aRMSE). These metrics, which reflect the deviation between the estimated and true abundances, are defined as follows:

$$\text{RMSE}(\hat{\mathbf{a}}_{pn}, \mathbf{a}_{pn}) = \sqrt{\frac{1}{N}\sum_{n=1}^{N} \|\hat{\mathbf{a}}_{pn} - \mathbf{a}_{pn}\|_2^2} \tag{27}$$

$$\text{aRMSE} = \frac{1}{P}\sum_{p=1}^{P} \text{RMSE}(\hat{\mathbf{a}}_{pn}, \mathbf{a}_{pn}) \tag{28}$$

where $\hat{\mathbf{a}}_{pn}$ and $\mathbf{a}_{pn}$ denote the estimated abundance and true abundance, respectively. Additionally, to evaluate the similarity between the true endmembers and those estimated by the proposed method, the SAD is defined by

$$\text{SAD}(\hat{\mathbf{e}}_p, \mathbf{e}_p) = \arccos\left(\frac{\hat{\mathbf{e}}_p^{\top}\mathbf{e}_p}{\|\hat{\mathbf{e}}_p\|_2\|\mathbf{e}_p\|_2}\right) \tag{29}$$

where $\hat{\mathbf{e}}_p$ and $\mathbf{e}_p$ denote the extracted and reference endmember, respectively. Its mean, referred to as aSAD, is calculated as an additional evaluation criterion

$$\text{aSAD} = \frac{1}{P}\sum_{p=1}^{P} \text{SAD}(\hat{\mathbf{e}}_p, \mathbf{e}_p). \tag{30}$$

C. Experiments on the Synthetic Dataset

Table III presents a detailed comparison of key performance metrics across different models on the synthetic dataset. These metrics include RMSE and aRMSE for abundance estimation accuracy, as well as SAD and aSAD for evaluating endmember similarity. Fig. 7 shows the abundance maps extracted by each method, highlighting differences in their spatial distributions. It can be observed that the FCLSU and CyCU-Net models performed poorly in both abundance estimation and EE, likely due to their limited ability to handle complex data features. When accounting for endmember variability, the PGMSU improved abundance unmixing, yielding more accurate estimation results. However, these models struggled to fully incorporate spatial information, which hindered further performance gains. The DGMCNN model, which integrates spatial information, showed unique strengths in abundance estimation for certain endmembers, underscoring the importance of spatial data in enhancing unmixing accuracy. Among these models, HUMSCAN demonstrated notable advantages by balancing endmember variability and spatial information, outperforming

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TABLE III

QUANTITATIVE RESULTS FOR THE SIM1 DATASET. ALL METRICS ARE AVERAGED OVER TEN INDEPENDENT RUNS. RMSE FOR EACH MATERIAL, ALONG WITH aRMSE, SAD, AND aSAD, ARE REPORTED. THE BEST RESULTS ARE HIGHLIGHTED IN RED, FOLLOWED BY THE SECOND-BEST RESULTS IN BLUE (SCALED BY 10^{-2})

Methods		FCLSU	CyCU-Net	PGMSU	DGMCNN	HUMSCAN	w-DSM	MUNet	SFANet	MSSF-Net
RMSE	EM #1	17.89 ± 0.56	26.49 ± 6.99	20.40 ± 0.94	20.46 ± 1.58	16.18 ± 2.06	17.89	13.76 ± 1.27	16.89 ± 0.84	15.74 ± 0.39
	EM #2	19.76 ± 5.79	17.02 ± 1.87	16.56 ± 3.26	12.70 ± 0.99	10.83 ± 2.45	7.86	10.41 ± 3.01	10.68 ± 0.80	9.76 ± 1.43
	EM #3	19.80 ± 2.88	25.55 ± 2.77	16.05 ± 3.53	10.92 ± 2.61	9.63 ± 1.80	4.72	12.53 ± 2.52	12.21 ± 0.27	7.78 ± 1.66
	EM #4	12.59 ± 1.51	17.71 ± 1.95	13.61 ± 2.36	12.45 ± 1.45	9.85 ± 0.90	11.49	10.49 ± 1.90	9.56 ± 0.67	11.42 ± 1.29
	EM #5	17.25 ± 5.66	18.16 ± 3.71	12.21 ± 4.78	14.34 ± 1.58	13.57 ± 3.31	5.39	14.56 ± 3.62	14.98 ± 0.65	3.20 ± 0.81
aRMSE		17.46 ± 1.40	20.99 ± 1.85	15.77 ± 1.56	14.18 ± 0.65	12.01 ± 0.74	9.47	12.35 ± 0.81	12.86 ± 0.35	9.58 ± 0.22
SAD	EM #1	2.18 ± 0.10	20.06 ± 6.62	4.39 ± 0.57	10.73 ± 6.44	1.37 ± 0.92	–	1.61 ± 0.80	3.13 ± 0.08	1.16 ± 0.09
	EM #2	23.78 ± 3.64	32.52 ± 4.66	23.48 ± 6.54	27.57 ± 1.66	19.66 ± 2.41	–	21.20 ± 6.91	19.18 ± 1.99	16.90 ± 1.15
	EM #3	24.87 ± 2.88	32.60 ± 5.29	22.62 ± 4.44	27.84 ± 4.82	33.78 ± 1.63	–	29.23 ± 1.59	29.76 ± 1.62	31.50 ± 7.23
	EM #4	4.07 ± 0.57	18.27 ± 9.49	3.80 ± 0.69	7.97 ± 1.26	3.36 ± 0.13	–	2.96 ± 0.52	2.83 ± 0.03	3.67 ± 0.06
	EM #5	43.51 ± 7.68	51.18 ± 6.66	36.90 ± 7.19	51.74 ± 7.13	43.47 ± 4.63	–	43.79 ± 10.50	34.38 ± 5.87	27.54 ± 2.14
aSAD		19.68 ± 2.29	30.93 ± 2.31	18.24 ± 2.47	25.17 ± 3.58	20.33 ± 0.42	–	19.76 ± 2.98	17.85 ± 1.44	16.16 ± 1.40

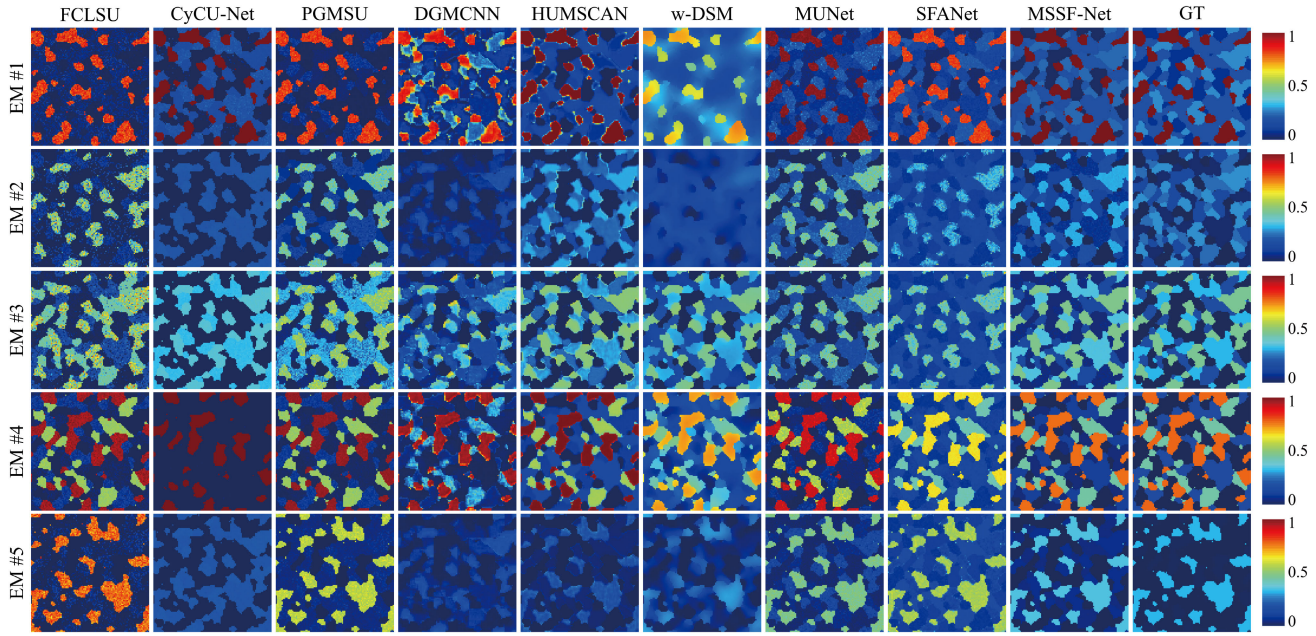


Fig. 7. Comparison of abundance maps for the SIM1 dataset, showing the abundance distribution differences of five materials under different unmixing algorithms.

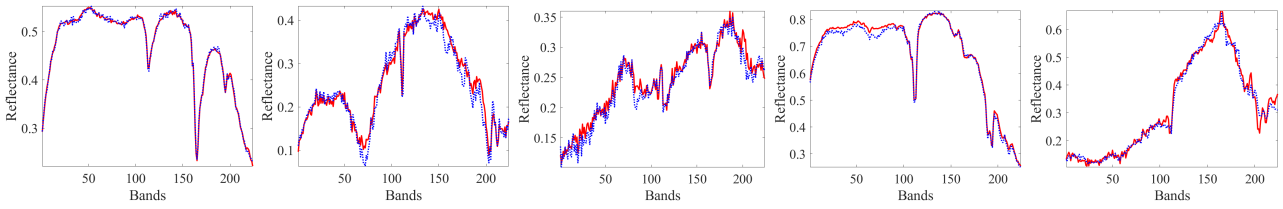


Fig. 8. Endmember comparison plots for the SIM1 dataset: the MSSF-Net (blue dotted line) and the corresponding ground truth (red curve). (a) EM #1. (b) EM #2. (c) EM #3. (d) EM #4. (e) EM #5.

others that focused on only one aspect. Nonetheless, all these models were limited to single-modal data processing, which prevented them from fully capturing the data's complexity, leading to suboptimal performance on the synthetic dataset.

In contrast, the MUNet model, through the integration of multimodal data, overcame the limitations of single-modal approaches and delivered superior abundance estimation. This demonstrates the significant potential of multimodal data processing in improving unmixing performance. w-DSM is a semi-supervised traditional method that requires manually preset endmembers as prior information. Additionally, it relies solely on LiDAR data for spatial regularization without integrating multisource features, resulting in suboptimal

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TABLE IV

QUANTITATIVE RESULTS FOR THE HOUSTON DATASET. ALL METRICS ARE AVERAGED OVER TEN INDEPENDENT RUNS. RMSE FOR EACH MATERIAL, ALONG WITH aRMSE, SAD, AND aSAD, ARE REPORTED. THE BEST RESULTS ARE HIGHLIGHTED IN RED, FOLLOWED BY THE SECOND-BEST RESULTS IN BLUE (SCALED BY 10^{-2})

Methods		FCLSU	CyCU-Net	PGMSU	DGMCNN	HUMSCAN	w-DSM	MUNet	SFANet	MSSF-Net
RMSE	Parking Lot1	15.75 ± 1.47	14.40 ± 3.40	8.32 ± 0.40	8.40 ± 0.54	7.48 ± 0.45	5.33	21.08 ± 10.16	21.45 ± 3.25	5.29 ± 0.79
	Parking Lot2	15.28 ± 1.47	22.06 ± 5.84	11.15 ± 0.57	16.39 ± 0.85	12.71 ± 0.30	12.27	22.07 ± 9.25	27.46 ± 3.07	5.66 ± 0.73
	Running Track	3.92 ± 0.84	23.56 ± 8.22	6.84 ± 1.47	8.97 ± 0..64	7.60 ± 0.73	4.07	11.34 ± 6.03	9.91 ± 4.04	6.45 ± 2.30
	Grass Healthy	10.17 ± 1.71	13.74 ± 6.10	4.70 ± 0.36	7.02 ± 0.36	5.31 ± 0.45	11.54	8.50 ± 0.79	8.30 ± 0.44	3.98 ± 0.55
aRMSE		11.28 ± 0.59	18.44 ± 4.01	7.75 ± 0.48	10.19 ± 3.70	8.27 ± 2.76	8.30	15.74 ± 9.57	16.78 ± 2.41	5.35 ± 0.88
SAD	Parking Lot1	19.16 ± 0.84	19.27 ± 8.18	1.61 ± 0.31	4.25 ± 0.29	1.95 ± 0.04	–	10.25 ± 9.84	0.95 ± 0.17	0.88 ± 0.24
	Parking Lot2	22.49 ± 3.42	34.45 ± 13.68	4.46 ± 0.21	11.69 ± 0.46	7.98 ± 0.29	–	4.80 ± 2.36	10.95 ± 1.47	2.16 ± 0.28
	Running Track	11.68 ± 4.46	24.65 ± 23.88	7.82 ± 0.18	11.65 ± 0.63	8.21 ± 0.17	–	12.87 ± 2.95	7.74 ± 3.92	13.11 ± 1.13
	Grass Healthy	15.21 ± 0.53	16.14 ± 9.79	4.99 ± 0.38	8.01 ± 0.63	5.42 ± 0.06	–	5.61 ± 0.68	7.03 ± 0.31	2.15 ± 0.67
aSAD		17.14 ± 1.26	23.63 ± 9.11	4.72 ± 0.17	8.90 ± 3.12	5.89 ± 2.83	–	8.36 ± 6.24	6.67 ± 1.05	4.55 ± 0.41

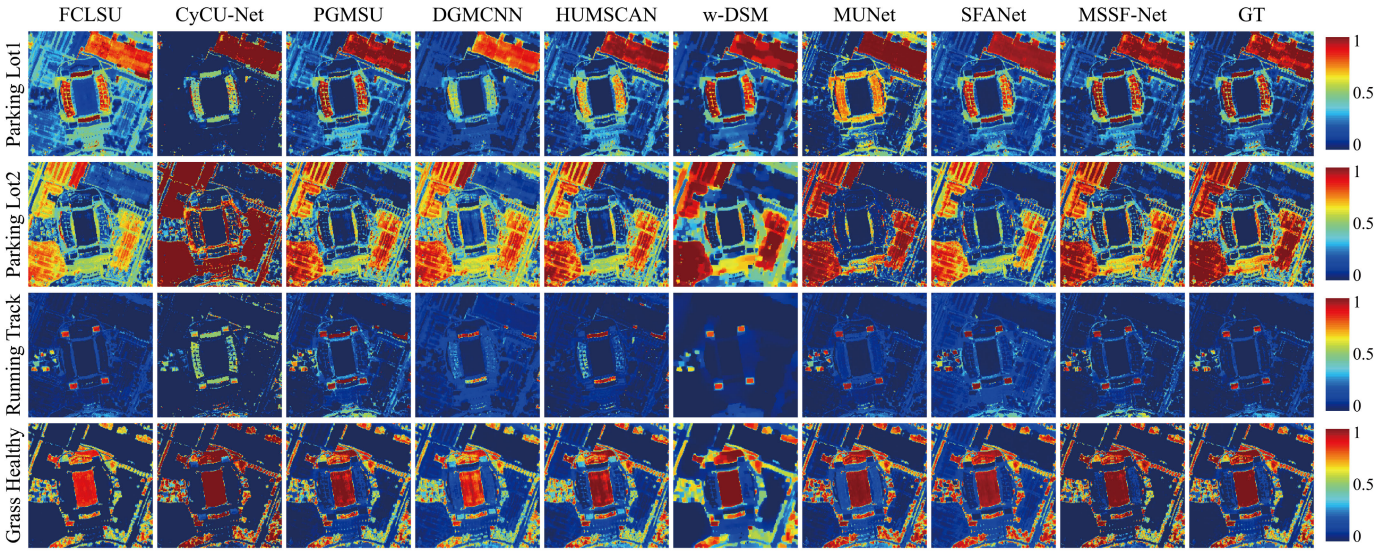


Fig. 9. Comparison of abundance maps for the Houston dataset, showing the abundance distribution differences of four materials under various unmixing algorithms.

TABLE V

QUANTITATIVE RESULTS FOR THE MUFFLE DATASET. ALL METRICS ARE AVERAGED OVER TEN INDEPENDENT RUNS. RMSE FOR EACH MATERIAL, ALONG WITH aRMSE, SAD, AND aSAD, ARE REPORTED. THE BEST RESULTS ARE HIGHLIGHTED IN RED, FOLLOWED BY THE SECOND-BEST RESULTS IN BLUE (SCALED BY 10^{-2})

Methods		FCLSU	CyCU-Net	PGMSU	DGMCNN	HUMSCAN	w-DSM	MUNet	SFANet	MSSF-Net
RMSE	Roof	22.78 ± 1.85	20.61 ± 4.31	14.84 ± 0.78	21.29 ± 1.58	18.31 ± 1.06	13.51	22.75 ± 8.55	16.68 ± 2.77	14.36 ± 0.32
	Grass	27.32 ± 2.40	24.42 ± 2.86	27.92 ± 1.17	30.58 ± 1.18	29.69 ± 2.17	21.57	25.78 ± 3.34	25.61 ± 3.67	22.53 ± 0.83
	Tree	24.25 ± 1.14	27.90 ± 3.39	27.41 ± 1.23	26.65 ± 1.85	29.45 ± 1.67	14.62	20.30 ± 2.27	20.77 ± 1.81	16.69 ± 0.90
	Shadow	31.92 ± 1.02	15.49 ± 4.03	4.88 ± 0.52	13.11 ± 1.79	6.32 ± 1.16	15.05	21.74 ± 6.74	13.95 ± 3.69	11.44 ± 0.77
	Asphalt	23.96 ± 1.09	28.76 ± 3.88	23.55 ± 1.05	23.17 ± 0.70	24.84 ± 0.92	13.62	17.09 ± 4.01	24.21 ± 0.94	23.29 ± 0.39
aRMSE		26.05 ± 1.22	23.44 ± 1.49	19.72 ± 0.49	22.96 ± 0.83	21.72 ± 0.82	15.67	21.53 ± 3.85	20.24 ± 2.51	17.66 ± 0.15
SAD	Roof	7.11 ± 0.01	2.81 ± 0.89	0.79 ± 0.08	12.22 ± 2.43	0.84 ± 0.44	–	2.59 ± 3.00	2.23 ± 0.71	4.46 ± 1.04
	Grass	13.36 ± 0.16	3.14 ± 0.78	20.39 ± 0.36	14.37 ± 1.42	24.00 ± 2.54	–	4.76 ± 2.90	14.35 ± 2.83	8.33 ± 1.52
	Tree	12.30 ± 3.20	12.14 ± 9.02	5.25 ± 0.33	8.98 ± 0.78	3.49 ± 0.52	–	4.16 ± 0.45	7.33 ± 1.22	2.51 ± 0.65
	Shadow	9.88 ± 0.41	21.99 ± 24.50	2.37 ± 0.67	16.06 ± 3.06	2.04 ± 0.31	–	10.05 ± 4.01	4.19 ± 0.85	2.24 ± 0.20
	Asphalt	16.09 ± 0.01	25.82 ± 20.54	13.23 ± 0.71	16.92 ± 3.21	10.71 ± 2.38	–	7.72 ± 5.21	12.57 ± 5.94	6.33 ± 1.56
aSAD		11.75 ± 0.66	13.18 ± 6.44	8.40 ± 0.31	13.71 ± 1.41	8.22 ± 0.17	–	5.85 ± 1.90	8.13 ± 1.65	4.78 ± 0.49

performance. In contrast, the SFANet incorporates the ELAM module to exploit global correlations in multisource data, enabling spectral–spatial mixed attention extraction. With preliminary improvements in feature fusion methods, the SFANet achieves better experimental results, particularly showing positive changes in the EM #4 metric. The proposed MSSF-Net model, which captures both global and local interaction of multimodal spectral–spatial features, achieved the best

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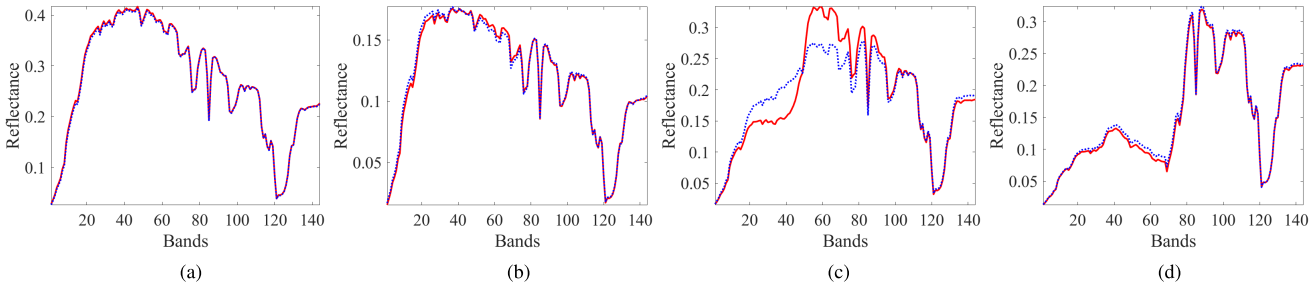


Fig. 10. Endmember comparison plots for the Houston dataset: the MSSF-Net (blue dotted line) and the corresponding ground truth (red curve). (a) Parking lot1. (b) Parking lot2. (c) Running track. (d) Grass healthy.

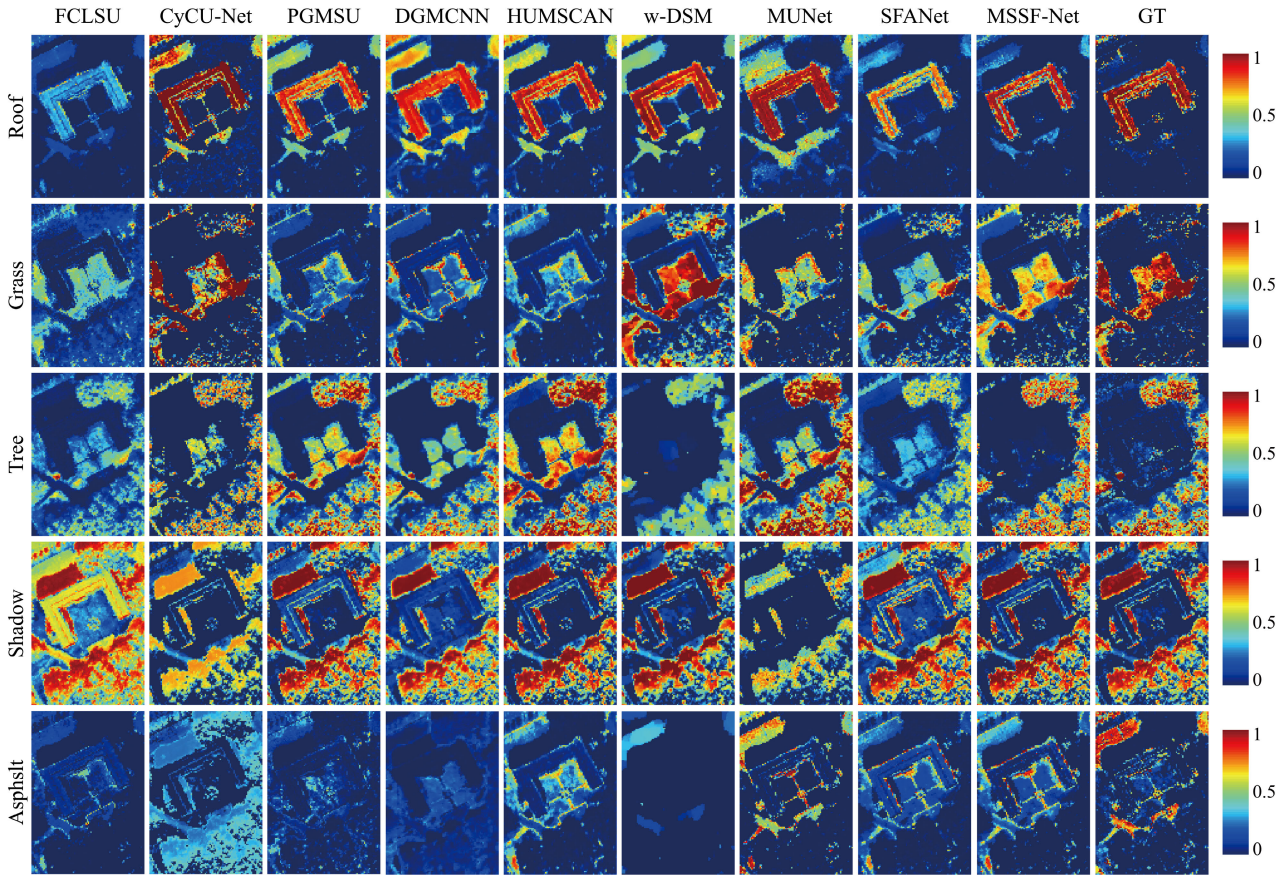


Fig. 11. Comparison of abundance maps for the Muffle dataset, showing the abundance distribution differences of five materials under different unmixing algorithms.

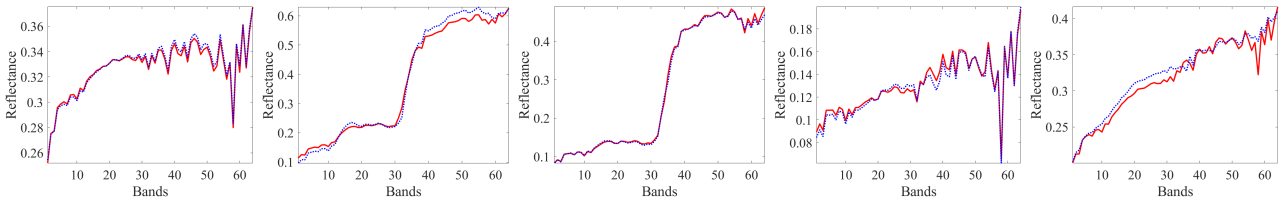


Fig. 12. Endmember comparison plots for the Muffle dataset: the MSSF-Net (blue dotted line) and the corresponding ground truth (red curve). (a) Roof. (b) Grass. (c) Tree. (d) Shadow. (e) Asphalt.

performance on the SIM1 dataset, confirming its superiority in HU tasks. Fig. 8 shows a comparison of the estimated endmembers from the proposed MSSF-Net with the corresponding ground truth, highlighting its excellent EE capabilities.

D. Experiments on the Houston Dataset

Table IV presents the quantitative results for the endmembers and abundance estimates obtained by different methods on the Houston2013 dataset, using RMSE, aRMSE, SAD,

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TABLE VI
ABLATION ANALYSIS OF THE MSSF-NET ON THE SIM1 DATASET WITH DIFFERENT MODULE COMBINATIONS (10^{-2})

Module				aRMSE	aSAD
3×3 Conv	ECAM	CMF			
		LFI	GFI		
				12.35 ± 0.81	19.76 ± 2.98
✓				11.94 ± 0.50	19.24 ± 1.86
	✓			12.14 ± 0.20	17.51 ± 3.56
✓	✓			11.28 ± 0.30	19.31 ± 1.99
		✓		11.84 ± 0.31	18.01 ± 1.28
			✓	11.92 ± 0.51	18.77 ± 1.78
		✓	✓	11.62 ± 0.42	17.56 ± 1.16
✓		✓	✓	10.96 ± 0.57	17.16 ± 1.25
	✓	✓	✓	10.75 ± 0.39	17.34 ± 1.98
✓	✓	✓		9.96 ± 0.27	17.13 ± 1.73
✓	✓		✓	10.14 ± 0.34	17.30 ± 1.33
✓	✓	✓	✓	9.58 ± 0.22	16.16 ± 1.40

and aSAD metrics. We observe that the PGMSU shows significant improvements in EE compared to both traditional methods like FCLSU and advanced methods like CyCU-Net. These improvements are attributed to their ability to account for endmember variability, underscoring the importance of considering dynamic changes in endmembers. The DGMCNN model stands out for its integration of spatial information with spectral analysis, leading to substantial performance gains over CyCU-Net in both EE and abundance estimation. This highlights the critical importance of spatial information in enhancing HU accuracy, particularly in complex scenes. Similarly, the HUMSCAN model, which combines both endmember variability and spatial information, further improves unmixing accuracy for specific endmembers compared to simpler models. These results emphasize the necessity of addressing multiple factors comprehensively and integrating diverse information sources to enhance the accuracy and reliability of HU. Meanwhile, both MUNet and SFANet effectively utilize multimodal data and show improvements in experimental results. However, their standard deviation values suggest certain limitations in robustness compared to the proposed MSSF-Net. This further highlights the importance of ensuring model stability while improving performance.

The proposed MSSF-Net model demonstrates exceptional performance, achieving substantial improvements of 30.97% in aRMSE and 3.6% in aSAD compared to the second-best model, PGMSU. These results are attributed to its effective utilization of multimodal data and seamless integration of spectral and spatial information, enabling efficient feature extraction. As shown in Fig. 9, FCLSU and CyCU-Net perform poorly in the unmixing task, particularly for Parking Lot1 and Parking Lot2. In contrast, the PGMSU effectively mitigates the effects of endmember variability, yielding visual results significantly closer to the ground truth. Among multimodal unmixing methods, MUNet and SFANet generate relatively smoother abundance maps. As expected, the abundance maps estimated by MSSF-Net closely align with the ground truth, visually confirming its superior unmixing capabilities. Furthermore, Fig. 10 illustrates the consistency between the endmember spectra extracted by the MSSF-Net and the real endmembers, showcasing remarkable alignment in both shape and trends. This further validates the precision and reliability of the MSSF-Net in EE.

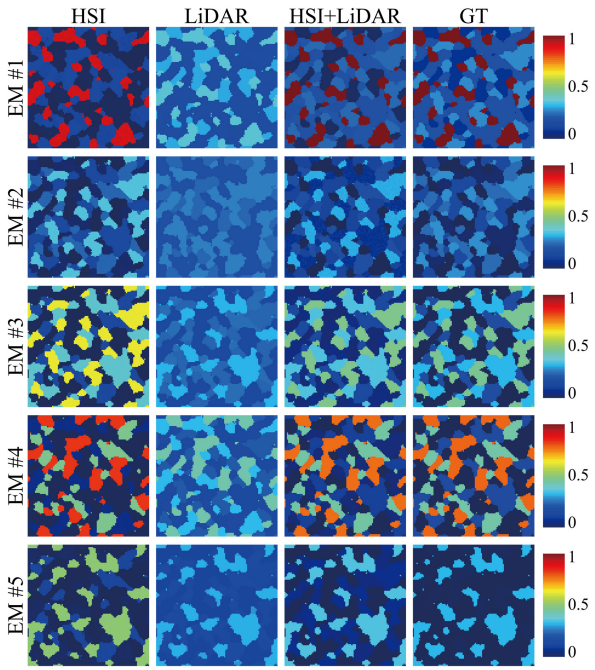


Fig. 13. Ablation study comparing abundance maps generated by the proposed MSSF-Net using single-modal and multimodal data.

E. Experiments on the Muffle Dataset

Table V presents the quantitative results of various HU methods on the Muffle dataset. It is clear that the Muffle dataset is more complex than the Houston dataset, primarily due to endmember variability, including subtle spectral differences between grass and tree, which significantly complicate the construction of an accurate spectral library. The results indicate that the FCLSU and CyCU-Net exhibit inadequate unmixing performance and significantly compromised stability when applied to such realistic and highly complex scenarios. Although the PGMSU and HUMSCAN account for endmember variability, they exhibit suboptimal performance on the grass and tree endmembers. In this challenging context, the proposed MSSF-Net demonstrates remarkable superiority, particularly in extracting the tree endmember, achieving notable improvements of 17.78% in RMSE and 28.08% in SAD compared to the second-best MUNet and HUMSCAN models, respectively. As illustrated in Fig. 11, increasingly complex scenes pose significant challenges, often resulting in higher error rates and more difficult interpretation of abundance maps. Nevertheless, the MSSF-Net accurately captures the shape characteristics of each material, closely matching the ground truth. Specifically, it achieves remarkable accuracy in estimating tree abundances, with minimal interference from other materials, such as grass. This highlights the efficiency and precision of the MSSF-Net in distinguishing complex land-cover features and handling challenging scenarios. However, the MSSF-Net encounters difficulties in extracting the spectral signature of the Asphalt endmember. Fig. 12

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TABLE VII
QUANTITATIVE RESULTS OF FEATURES EXTRACTED FROM DIFFERENT MODALITIES ON THE HOUSTON AND MUFFLE DATASETS USING THE PROPOSED MSSF-NET (10^{-2})

Dataset		RMSE					aRMSE
		Parking Lot1	Parking Lot2	Running Track	Grass Healthy	–	
Houston	HSI	7.61 ± 1.11	7.77 ± 0.96	8.46 ± 0.92	5.89 ± 1.35	–	7.43 ± 0.87
	LiDAR	29.77 ± 1.79	38.69 ± 1.82	19.84 ± 1.45	34.50 ± 0.42	–	30.70 ± 7.18
	HSI + LiDAR	5.29 ± 0.79	5.66 ± 0.73	6.45 ± 2.30	3.98 ± 0.55	–	5.35 ± 0.88
		Roof	Grass	Tree	Shadow	Asphalt	
Muffle	HSI	17.73 ± 1.31	24.88 ± 2.99	20.42 ± 2.83	12.64 ± 1.26	25.45 ± 1.11	20.22 ± 5.94
	LiDAR	29.83 ± 1.52	33.85 ± 0.46	28.71 ± 0.87	36.99 ± 0.48	25.99 ± 0.40	31.07 ± 3.98
	HSI + LiDAR	14.36 ± 0.32	22.53 ± 0.83	16.69 ± 0.90	11.44 ± 0.77	23.29 ± 0.39	17.66 ± 0.15

TABLE VIII
RUNTIME OBTAINED BY AVERAGING THE TIME CONSUMED OVER TEN INDEPENDENT RUNS

Method	Runtime (s)		
	SIM1	Houston	Muffle
FCLSU	0.26	0.42	0.14
CyCU-Net	1.95	1.38	1.19
PGMSU	11.18	22.79	26.12
DGMCNN	11.26	76.69	25.48
HUMSCAN	15.76	26.63	15.99
w-DSM	4.69	10.22	3.78
MUNet	18.52	15.83	13.85
SFANet	15.18	56.67	20.49
MSSF-Net	16.79	37.38	8.75

visualizes the endmember spectra, revealing noticeable deviations from the reference spectrum, which impact the accuracy of subsequent abundance estimates. While MSSF-Net does not achieve the lowest RMSE and SAD values for all individual endmembers, its comprehensive performance, as reflected in aRMSE and aSAD metrics, demonstrates consistently more accurate and stable abundance and endmember estimation results compared to competing models.

F. Ablation Experiments

Through carefully designed ablation experiments, we validated the critical contributions of the ECAM mechanism, the CMF module, and spatial information extraction in improving the unmixing performance of the MSSF-Net. When the CMF module was removed, the original fusion mechanism of the baseline model was retained, where the feature outputs from both modalities were directly fused using an element-wise weighted multiplication strategy. As shown in Table VI, the unmixing accuracy significantly declined when key components such as the 3×3 convolution layer in the HSI branch, the ECAM in the LiDAR branch, and the CMF module were removed. These results highlight the indispensable role of these components in effectively extracting and integrating multisource information. Their inclusion fosters robust feature representation and enhances multimodal information fusion, ultimately improving unmixing accuracy. Specifically, the 3×3 convolution layer in the HSI branch captures spatial relationships, significantly enhancing the accuracy of abundance estimation and endmember identification. Meanwhile, the ECAM mechanism in the LiDAR branch enriches the representation of elevation variations, providing strong support for endmember identification. Additionally, the LFI and GFI components in the CMF module enable seamless fusion of HSI and LiDAR features, outperforming single-fusion modules. This integrated fusion strategy fully exploits the complementary advantages of both modalities, offering comprehensive support for HU tasks and demonstrating exceptional robustness. These findings underscore the importance of these components and demonstrate the significant potential of multisource information fusion for unmixing in complex environments.

To thoroughly investigate the impact of different modalities on unmixing performance, we conducted ablation experiments using three input configurations: HSI data alone, LiDAR data alone, and the combination of HSI and LiDAR data. Fig. 13 presents a comparison of the abundance maps extracted by the HSI feature extraction module, the LiDAR feature extraction module, and the MSSF-Net across these three modalities on the SIM1 dataset, with each row corresponding to a specific endmember. As shown in the first column of Fig. 13, relying solely on HSI data results in inaccurate abundance maps, particularly for EM #2 and EM #3. In contrast, incorporating LiDAR data, as illustrated in the second column, significantly improves the accuracy of the extracted abundance features, highlighting the complementary relationship between HSI and LiDAR data during fusion. Table VII provides quantitative results for the Houston and Muffle datasets, further substantiating the complementary roles of HSI and LiDAR data. While HSI features alone struggle to capture complex scene characteristics, their integration with LiDAR features greatly enhances feature extraction accuracy. These findings, supported by both qualitative and quantitative evidence, confirm the superior unmixing performance of the proposed MSSF-Net and its effectiveness in leveraging multisource information fusion.

G. Computational Cost

In this subsection, we systematically evaluate the computational efficiency of various comparison methods across different datasets, focusing specifically on their runtimes. All methods were tested under a standardized hardware environment to ensure fairness and accuracy in the evaluation.

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As shown in Table VIII, AE-based models, while demonstrating strong performance, exhibit longer computation times due to their complex architectures and large parameter sets. In contrast, traditional physical methods leverage their simplicity, resulting in faster processing speeds.

For the proposed MSSF-Net, one key factor influencing runtime is the batch size during training. The batch size directly affects the number of iterations required to process the dataset per training epoch. Specifically, smaller batch sizes may enhance model generalization and stability but require more iterations to complete an epoch, thereby increasing overall computation time. Conversely, larger batch sizes reduce the number of iterations but may compromise convergence speed and optimization. Striking an appropriate balance between efficiency and performance is therefore critical in practice. Despite these considerations, the computational cost of the MSSF-Net remains acceptable and aligns with its performance advantages.

V. Conclusion

This work presented MSSF-Net, a novel multimodal spectral unmixing network designed to integrate spatial and spectral features from HSI and LiDAR data. By combining the spatial analysis capabilities of 2-D convolution with the ECAM mechanism for spectral precision, the MSSF-Net effectively captures both fine spatial and spectral details. The CMF module further enhances cross-modal interaction, improving both accuracy and robustness. Experimental results demonstrate that the MSSF-Net outperforms existing methods, achieving outstanding unmixing results on synthetic datasets and excellent performance in real-world applications. In future work, we plan to develop unmixing networks with enhanced generalization capabilities and extend the MSSF-Net to address both endmember variability and nonlinear unmixing, particularly for complex environments and diverse surface conditions.

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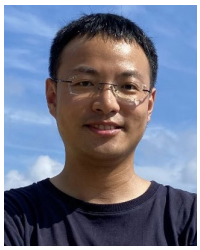
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